

Aubrey David

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Education

The University of Georgia, BS Computer Systems Engineering – Athens, GA

May 2026

GPA: 3.26

Experience

Software Engineering Intern, Cox Automotive – Atlanta, GA

May – August 2025

- Contributed to the development of internal tools and APIs, improving team efficiency and code quality across services.
- Competed in a company-wide hackathon, collaborating with engineers and product employees to prototype an AI-powered review summary for customers.
- Developed a time-saving application to streamline workflows for product employees, projected to reduce manual effort by up to 5 hours per week.
- Worked in an Agile team environment, participating in daily standups, sprint planning, and code reviews while gaining exposure to enterprise-scale development practices.

Mentor, Redcoat Marching Band, University of Georgia – Athens, GA

August 2022 – May 2025

- Collaborated with a team of 400+ members to perform intricate field routines during football games and events. Consistently demonstrated precision and teamwork, contributing to a positive UGA gameday experience.
- Balanced 20+ hours per week of rehearsals and performances alongside a full academic course load, developing time management and organizational skills.

Sales Associate, Belk, Inc. – Evans, GA

June 2022 – July 2023

- Delivered customer service and exceeded monthly credit sign-up goals (up to 20%).

Projects

Smart Instrument Tuner

- Developed a smart instrument tuner using force, vibration, and humidity sensors to provide accurate pitch detection and environmental monitoring. Implemented all signal processing, enabling precise pitch detection within ± 1 Hz.
- Resolved noise interference in audio signals by implementing digital filtering and optimizing a fast-fourier transform algorithm. Enhanced signal clarity, ensuring consistent performance in noisy environments.

Android App RP4 Motor Control

- Architected a robust wireless interface between a native Android (Kotlin) app and an embedded C server on a Raspberry Pi using a custom RFCOMM Bluetooth protocol and state machine parser to ensure stable, low-latency data transmission.
- Streamlined device lifecycle management by developing systemd services to automate Bluetooth pairing, achieving a true headless user experience similar to consumer "plug-and-play" instrument accessories.

AES Cryptographic Accelerator

- Designed and synthesized a hardware-accelerated AES-128 encryption/decryption engine on an FPGA, interfaced with UART protocol, and delivered sub-millisecond, bit-accurate cryptographic processing for secure data handling.
- Optimized FPGA resource utilization by pipelining AES rounds, achieving low-latency throughput within strict LUT constraints.
- Built and executed self-checking Verilog testbenches using known AES vectors, validating every encryption and decryption stage for 100% functional correctness and cryptographic integrity.

Skills

Languages: Java, C/C++, Python, Verilog, Assembly, JavaScript, Bash, MATLAB, Kotlin

Software: Git, REST API, Linux/Unix Environment, GDB/GEF, React, Docker, AWS, Vivado, Codesys

Hardware: Circuit Analysis, Arduino, ESP32, Raspberry Pi, IoT, FPGA Nexys A7

Modeling: Multisim, FactoryIO, Tinkercad